

Context free grammar -

- Context free grammar is a formal grammar which is used to generate all possible strings in a given formal language.
- Context free grammar G can be defined by four tuples as:

$$G = (V, T, P, S)$$

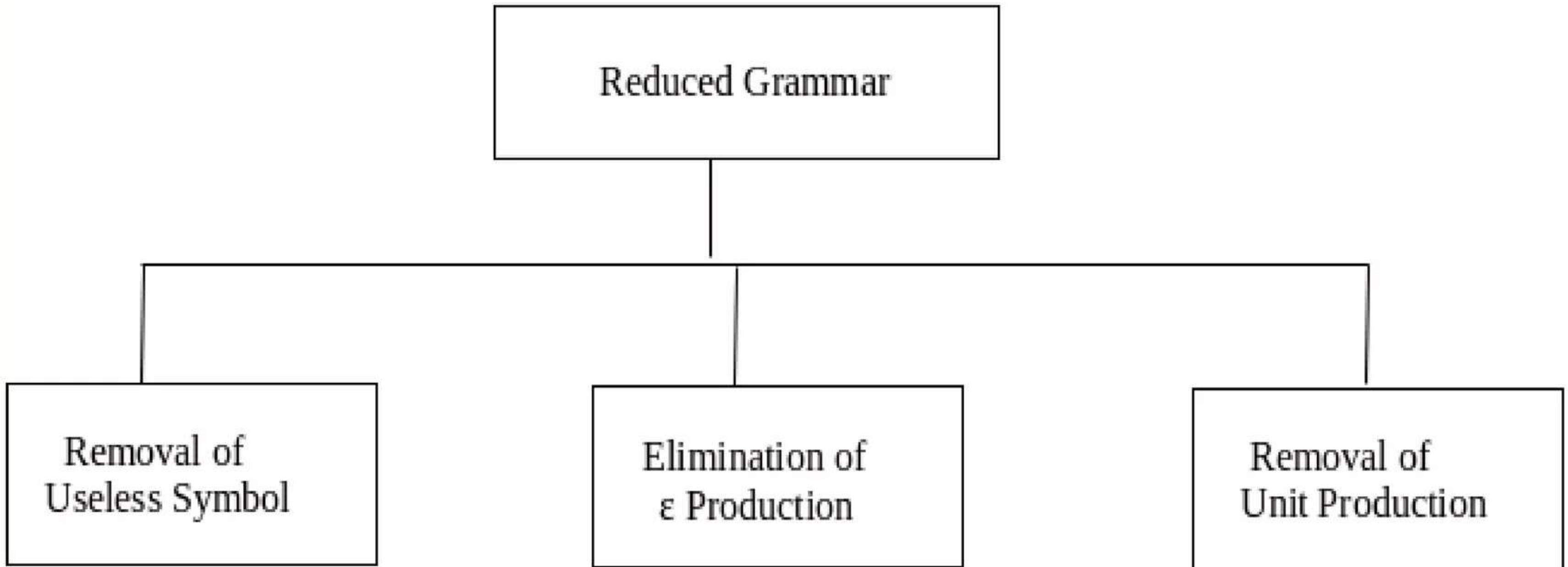
Where,

- G - describes the grammar
- T - describes a finite set of terminal symbols.
- V - describes a finite set of non-terminal symbols
- P - describes a set of production rules
- S - is the start symbol.

Simplification of CFG -

- All the grammar are not always optimized that means the grammar may consist of some **extra symbols**(non-terminal). Having **extra symbols**, unnecessary **increase the length of grammar**. Simplification of grammar means **reduction** of grammar by **removing useless symbols**. The properties of reduced grammar are given below:
 1. Each **variable** (i.e. non-terminal) and each **terminal** of G appears in the derivation of some word in L .
 2. There should not be any production as $X \rightarrow Y$ where X and Y are **non-terminal**.
 3. If ϵ is not in the language L then there need not to be the production $X \rightarrow \epsilon$.

Types of redundant productions in CFG -



Removal of Useless Symbols -

A **symbol** can be **useless** if it does not appear on the **right-hand side** of the **production rule** and does not take part in the derivation of any string. That symbol is known as **a useless symbol**. Similarly, a **variable** can be **useless** if it does not take part in the **derivation** of any string. That variable is known as **a useless variable**.

For Example:

1. $T \rightarrow aaB \mid abA \mid aaT$
2. $A \rightarrow aA$
3. $B \rightarrow ab \mid b$
4. $C \rightarrow ad$

The production $C \rightarrow ad$ is **useless**. So we will **eliminate** it.

Production $A \rightarrow aA$ is also **useless** because there is **no way** to **terminate** it.

❖ Collectively we can rewrite the CFG with removed ϵ production as

1. $T \rightarrow aaB \mid abA \mid aaT$

2. $A \rightarrow aA$

3. $B \rightarrow ab \mid b$

Elimination of ϵ Production -

The **productions** of type $S \rightarrow \epsilon$ are called **ϵ productions**. These type of productions can only be **removed** from those grammars that do **not generate ϵ** .

Step 1: First find out all **nullable non-terminal variable** which derives ϵ .

Step 2: For each production $A \rightarrow a$, construct all production $A \rightarrow x$, where x is obtained from a by **removing one or more non-terminal** from step 1.

Step 3: Now combine the result of step 2 with the **original production** and **remove ϵ productions**.

Example:

1. $S \rightarrow XYX$

2. $X \rightarrow 0X \mid \varepsilon$

3. $Y \rightarrow 1Y \mid \varepsilon$

while removing ε production, we are deleting the rule $X \rightarrow \varepsilon$ and $Y \rightarrow \varepsilon$. To preserve the meaning of CFG we are actually placing ε at the right-hand side whenever X and Y have appeared.

$$S \rightarrow XYX$$

❖ If the first X at right-hand side is ε . Then

$$S \rightarrow YX$$

❖ Similarly if the last X in R.H.S. = ε . Then

$$S \rightarrow XY$$

❖ If Y and X are ε then,

$$S \rightarrow X$$

❖ If both X are replaced by ε

$$S \rightarrow Y$$

❖ Now,

$$S \rightarrow XY \mid YX \mid XX \mid X \mid Y$$

❖ Now let us consider

$$X \rightarrow 0X$$

❖ If we place ε at right-hand side for X then,

1. $X \rightarrow 0$

2. $X \rightarrow 0X \mid 0$

❖ Similarly,

$$Y \rightarrow 1Y \mid 1$$

❖ Collectively we can rewrite the CFG with removed ε production as

1. $S \rightarrow XY \mid YX \mid XX \mid X \mid Y$

2. $X \rightarrow 0X \mid 0$

3. $Y \rightarrow 1Y \mid 1$

Removing Unit Productions-

The **unit productions** are the productions in which one **non-terminal** gives another **non-terminal**. Use the following steps to remove unit production:

Step 1: To **remove** $X \rightarrow Y$, add production $X \rightarrow a$ to the grammar rule whenever $Y \rightarrow a$ occurs in the grammar.

Step 2: Now **delete** $X \rightarrow Y$ from the grammar.

Step 3: Repeat step 1 and step 2 until all unit productions are removed.

For example:

1. $S \rightarrow 0A \mid 1B \mid C$

2. $A \rightarrow 0S \mid 00$

3. $B \rightarrow 1 \mid A$

4. $C \rightarrow 01$

❖ $S \rightarrow C$ is a unit production.

$$S \rightarrow 0A \mid 1B \mid 01$$

❖ Similarly, $B \rightarrow A$ is also a unit production so we can modify it as

$$B \rightarrow 1 \mid 0S \mid 00$$

❖ Thus finally we can write CFG without unit production as

1. $S \rightarrow 0A \mid 1B \mid 01$

2. $A \rightarrow 0S \mid 00$

3. $B \rightarrow 1 \mid 0S \mid 00$

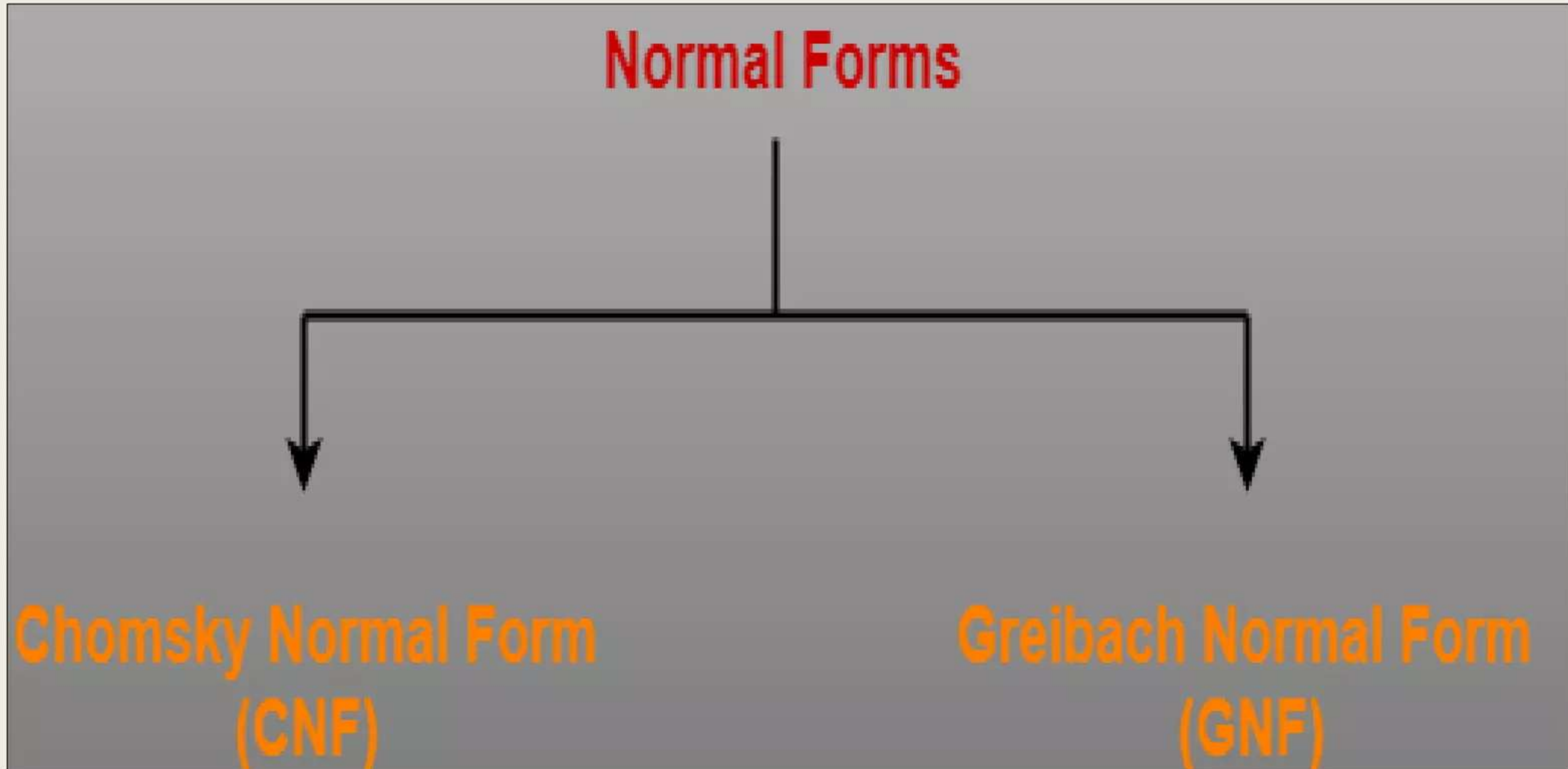
4. $C \rightarrow 01$

Normal Forms for CFG-

- Normalization is performed in order to standardize the grammar.
- By reducing the grammar, the grammar gets minimized but does not get standardized.
- This is because the RHS of productions have no specific format.
- In order to standardize the grammar, normalization is performed using normal forms.

TYPES OF NORMAL FORMS-

- The most frequently used normal forms are-



CHOMSKY's NORMAL FORM (CNF)-

- CNF stands for **Chomsky normal form**. A CFG(context free grammar) is in CNF(Chomsky normal form) if all production rules satisfy one of the following conditions:
 - **Start symbol** generating ϵ . For example, $A \rightarrow \epsilon$.
 - A **non-terminal** generating **two non-terminals**. For example, $S \rightarrow AB$.
 - A **non-terminal** generating **a terminal**. For example, $S \rightarrow a$.

For example:

1. $G1 = \{S \rightarrow AB, S \rightarrow c, A \rightarrow a, B \rightarrow b\}$
2. $G2 = \{S \rightarrow aA, A \rightarrow a, B \rightarrow c\}$

The **production rules** of Grammar **G1** **satisfy** the **rules** specified for **CNF**, so the grammar **G1** is in **CNF**. However, the **production rule** of Grammar **G2** **does not satisfy** the **rules** specified for **CNF** as $S \rightarrow aZ$ contains **terminal** followed by **non-terminal**. So the grammar **G2** is **not in CNF**.

Steps for convert a CFG into CNF-

Step 1: Eliminate **start symbol** from the RHS. If the **start symbol T** is at the **right-hand side** of any production, create a **new production** as:

$$S1 \rightarrow S$$

Where **S1** is the new **start symbol**.

Step 2: In the grammar, **remove the null, unit and useless** productions. You can refer to the [Simplification of CFG](#)

Step 3: Eliminate terminals from the RHS of the production if they exist with other non-terminals or terminals.

For example- production $S \rightarrow aA$ can be decomposed as:

$$S \rightarrow RA$$

$$R \rightarrow a$$

Step 4: Eliminate RHS with more than two non-terminals. For example, $S \rightarrow ASB$ can be decomposed as:

$$S \rightarrow RS$$

$$R \rightarrow AS$$

Example:

Convert the given CFG to CNF. Consider the given **grammar G1**:

1. $S \rightarrow a \mid aA \mid B$
2. $A \rightarrow aBB \mid \epsilon$
3. $B \rightarrow Aa \mid b$

Solution:

❖ We will **create** a new **production** $S1 \rightarrow S$, as the **start symbol** S appears on the **RHS**. The grammar will be:

1. $S1 \rightarrow S$
2. $S \rightarrow a \mid aA \mid B$
3. $A \rightarrow aBB \mid \epsilon$
4. $B \rightarrow Aa \mid b$

❖ As grammar **G1** contains $A \rightarrow \varepsilon$ null production, its removal from the grammar yields:

1. $S1 \rightarrow S$

2. $S \rightarrow a \mid aA \mid B$

3. $A \rightarrow aBB$

4. $B \rightarrow Aa \mid b \mid a$

❖ Now, as grammar **G1** contains Unit production $S \rightarrow B$, its removal yield:

1. $S1 \rightarrow S$

2. $S \rightarrow a \mid aA \mid Aa \mid b$

3. $A \rightarrow aBB$

4. $B \rightarrow Aa \mid b \mid a$

❖ Also **remove** the **unit production** $S1 \rightarrow S$, its removal from the grammar yields:

1. $S0 \rightarrow a \mid aA \mid Aa \mid b$

2. $S \rightarrow a \mid aA \mid Aa \mid b$

3. $A \rightarrow aBB$

4. $B \rightarrow Aa \mid b \mid a$

❖ In the production rule $S0 \rightarrow aA \mid Aa$, $S \rightarrow aA \mid Aa$, $A \rightarrow aBB$ and $B \rightarrow Aa$, **terminal** a exists on RHS **with non-terminals**. So we will **replace terminal a with X** :

1. $S0 \rightarrow a \mid XA \mid AX \mid b$

2. $S \rightarrow a \mid XA \mid AX \mid b$

3. $A \rightarrow XBB$

4. $B \rightarrow AX \mid b \mid a$

5. $X \rightarrow a$

❖ In the production rule $A \rightarrow XBB$, RHS has more than **two symbols**, **removing** it from grammar yield:

1. $S \rightarrow a \mid XA \mid AX \mid b$

2. $S \rightarrow a \mid XA \mid AX \mid b$

3. $A \rightarrow RB$

4. $B \rightarrow AX \mid b \mid a$

5. $X \rightarrow a$

6. $R \rightarrow XB$

Hence, for the given grammar, this is the required CNF.

Greibach Normal Form (GNF)-

- GNF stands for **Greibach normal form**. A CFG(context free grammar) is in GNF(Greibach normal form) if all the production rules satisfy one of the following conditions:
 - A **start symbol** generating ϵ . For example, $S \rightarrow \epsilon$.
 - A **non-terminal** generating a **terminal**. For example, $A \rightarrow a$.
 - A **non-terminal** generating a **terminal** which is followed by any **number of non-terminals**. For example, $S \rightarrow aASB$.

For example:

1. $G1 = \{S \rightarrow aAB \mid aB, A \rightarrow aA \mid a, B \rightarrow bB \mid b\}$
2. $G2 = \{S \rightarrow aAB \mid aB, A \rightarrow aA \mid \epsilon, B \rightarrow bB \mid \epsilon\}$

The **production rules** of Grammar **G1** **satisfy** the **rules** specified for **GNF**, so the grammar **G1** is **in GNF**. However, the **production rule** of Grammar **G2** **does not satisfy** the **rules** specified for **GNF** as $A \rightarrow \epsilon$ and $B \rightarrow \epsilon$ contains ϵ (only start symbol can generate ϵ). So the grammar **G2** is **not in GNF**.

Steps for converting CFG into GNF-

Step 1: Convert the grammar into CNF.

If the given **grammar is not in CNF**, convert it into CNF. You can refer the following topic to convert the CFG into CNF: Chomsky normal form

Step 2: If the **grammar** exists **left recursion**, **eliminate** it.

If the context free grammar contains **left recursion**, eliminate it. You can refer the following topic to eliminate left recursion: Left Recursion

Step 3: In the grammar, **convert** the given **production rule** into **GNF form**.

If any production rule in the grammar is not in GNF form, convert it.

Example:

1. $S \rightarrow XB \mid AA$
2. $A \rightarrow a \mid SA$
3. $B \rightarrow b$
4. $X \rightarrow a$

Solution:

- ❖ As the given **grammar** G is **already in CNF** and there is **no left recursion**, so we can skip step 1 and step 2 and directly go to step 3.
- ❖ The production rule $A \rightarrow SA$ is **not in GNF**, so we substitute $S \rightarrow XB \mid AA$ in the production rule $A \rightarrow SA$ as:

❖ The production rule $A \rightarrow SA$ is not in GNF, so we substitute $S \rightarrow XB \mid AA$ in the production rule $A \rightarrow SA$ as:

1. $S \rightarrow XB \mid AA$

2. $A \rightarrow a \mid XBA \mid AAA$

3. $B \rightarrow b$

4. $X \rightarrow a$

❖ The production rule $S \rightarrow XB$ and $B \rightarrow XBA$ is not in GNF, so we substitute $X \rightarrow a$ in the production rule $S \rightarrow XB$ and $B \rightarrow XBA$ as:

1. $S \rightarrow aB \mid AA$

2. $A \rightarrow a \mid aBA \mid AAA$

3. $B \rightarrow b$

4. $X \rightarrow a$

❖ Now we will **remove left recursion** ($A \rightarrow AAA$), we get:

1. $S \rightarrow aB \mid AA$

2. $A \rightarrow aC \mid aBAC$

3. $C \rightarrow AAC \mid \epsilon$

4. $B \rightarrow b$

5. $X \rightarrow a$

❖ Now we will **remove null production** $C \rightarrow \epsilon$, we get:

1. $S \rightarrow aB \mid AA$

2. $A \rightarrow aC \mid aBAC \mid a \mid aBA$

3. $C \rightarrow AAC \mid AA$

4. $B \rightarrow b$

5. $X \rightarrow a$

❖ The production rule $S \rightarrow AA$ is not in GNF, so we substitute $A \rightarrow aC \mid aBAC \mid a \mid aBA$ in production rule $S \rightarrow AA$ as:

1. $S \rightarrow aB \mid aCA \mid aBACA \mid aA \mid aBAA$

2. $A \rightarrow aC \mid aBAC \mid a \mid aBA$

3. $C \rightarrow AAC$

4. $C \rightarrow aCA \mid aBACA \mid aA \mid aBAA$

5. $B \rightarrow b$

6. $X \rightarrow a$

❖ The production rule $C \rightarrow AAC$ is not in GNF, so we substitute $A \rightarrow aC \mid aBAC \mid a \mid aBA$ in production rule $C \rightarrow AAC$ as:

1. $S \rightarrow aB \mid aCA \mid aBACA \mid aA \mid aBAA$
2. $A \rightarrow aC \mid aBAC \mid a \mid aBA$
3. $C \rightarrow aCAC \mid aBACAC \mid aAC \mid aBAAC$
4. $C \rightarrow aCA \mid aBACA \mid aA \mid aBAA$
5. $B \rightarrow b$
6. $X \rightarrow a$

Hence, this is the GNF form for the grammar G.